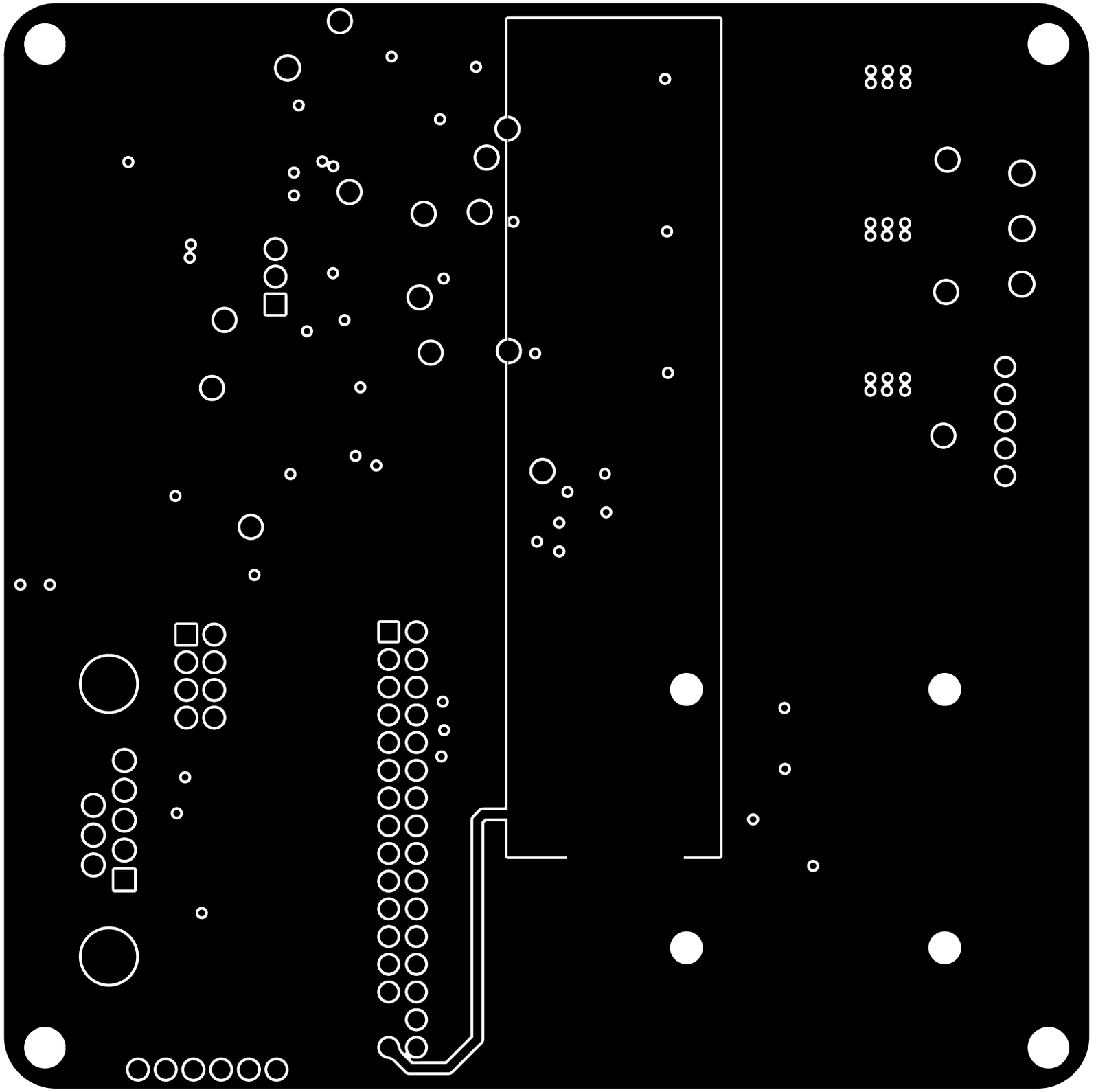
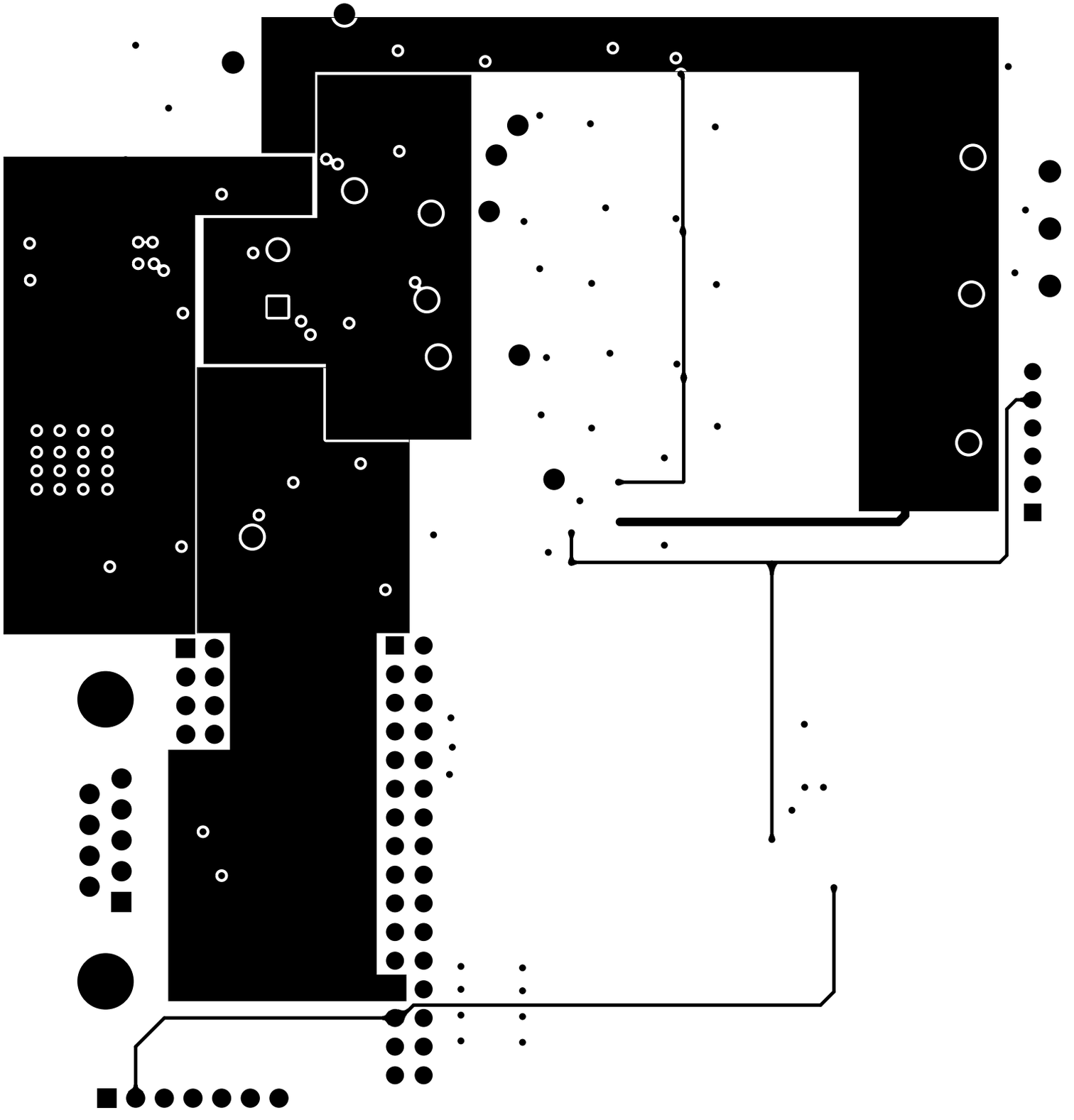
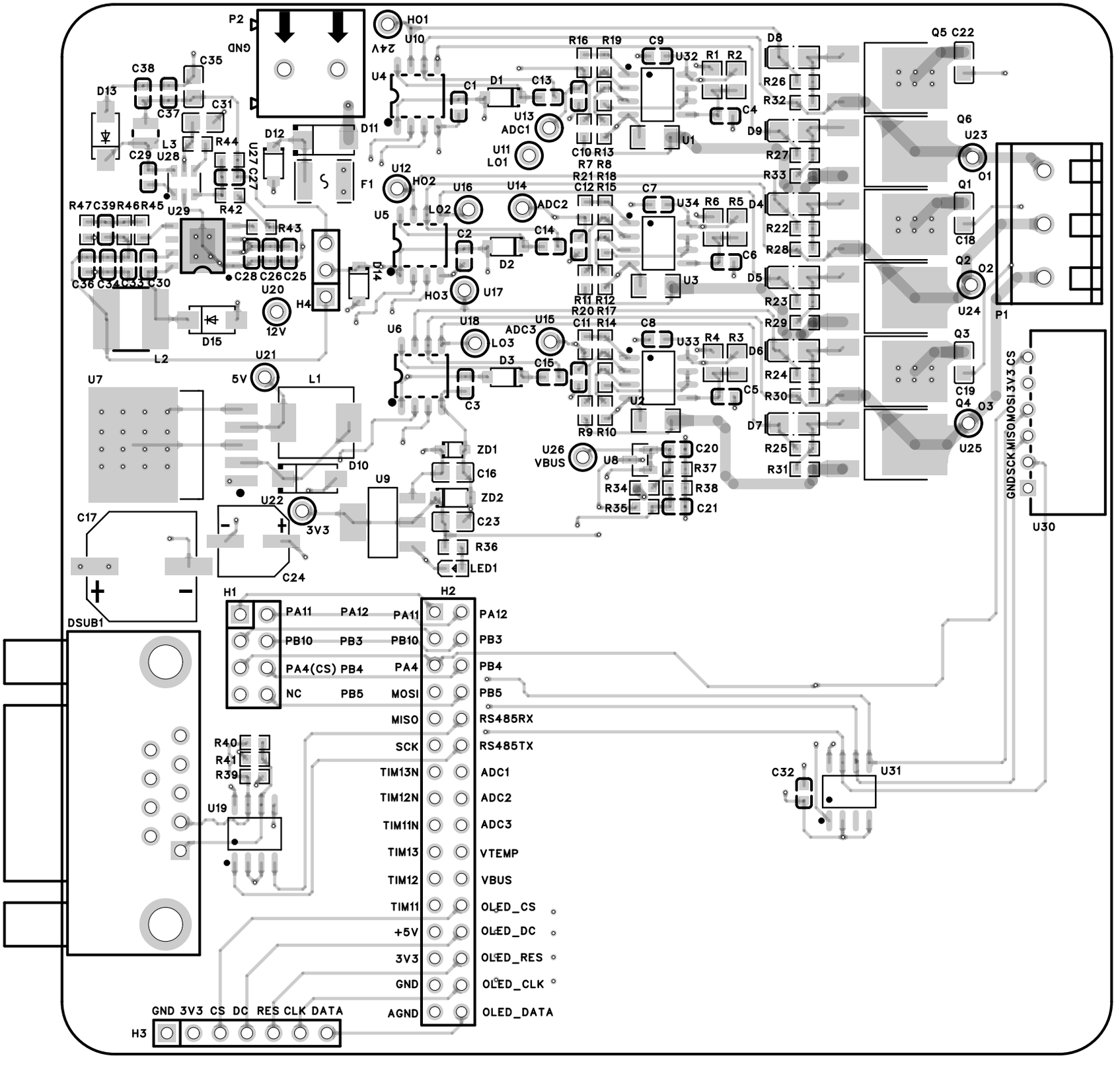





P2
GND
HO1
24V
U10
U4
C1
D1
C13
U13
ADC1
U11
LO1
R16
R19
C9
U32
R1
R2
C4
U1
D8
R26
R32
D9
R27
R33
D4
R22
R28
D5
R23
R29
D6
R24
R30
D7
R25
R31
Q5
C22
Q6
U23
O1
Q1
C18
Q2
O2
U24
Q3
C19
Q4
O3
U25
P1
GND
SCK
MISO
MOSI
3V3
CS
U30
D13
C38
C35
C37
C31
L3
C29
U28
R44
U27
C27
D12
D11
F1
U12
HO2
U16
LO2
U14
ADC2
U5
C2
D2
C14
R21
R18
C12
R15
C7
U34
R6
R5
C6
U3
R11
R12
R20
R17
C11
R14
C8
U33
R4
R3
C5
U2
R9
R10
U26
VBUS
U8
C20
R37
R34
R38
R35
C21
R47
C39
R46
R45
U29
R42
R43
C28
C26
C25
C36
C34
C33
C30
U20
12V
H4
D14
D15
L2
HO3
U17
U18
LO3
U15
ADC3
U6
D3
C15
C3
U21
5V
L1
U7
D10
U9
ZD1
C16
ZD2
C23
R36
LED1
U22
3V3
C17
C24
DSUB1
H1
H2
PA11
PA12
PB10
PB3
PA4(CS)
PB4
NC
PB5
MOSI
MISO
SCK
TIM13N
TIM12N
TIM11N
TIM13
TIM12
TIM11
+5V
3V3
GND
AGND
RS485RX
RS485TX
ADC1
ADC2
ADC3
VTEMP
VBUS
OLED_CS
OLED_DC
OLED_RES
OLED_CLK
OLED_DATA
R40
R41
R39
U19
C32
U31
H3
GND 3V3 CS DC RES CLK DATA

